# The Impact of Device Type and Payment Method on Cart Abandonment: An Experimental Analysis

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2026-03-22

## 1. Research Question

**How do device type and preferred payment method impact the cart abandonment rate of users?**

## 2. Methodology

To answer this research question, this study employs an **Experimental Design and A/B Testing** approach. Specifically, a **Two-Way ANOVA** (Analysis of Variance) is utilized to evaluate the main effects and the interaction effect of device type and payment method on the cart abandonment rate [2].

This quantitative method is chosen because it allows for the examination of causal inference between categorical independent variables (device type, payment method) and a continuous dependent variable (abandonment rate). The underlying mathematical model for this analysis can be expressed as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \epsilon$$

*(Where: $Y$ is cart_abandonment_rate, $X_1$ is device_type, $X_2$ is preferred_payment_method, and $\epsilon$ is the random error term)* [3].

### Data Import and Preparation

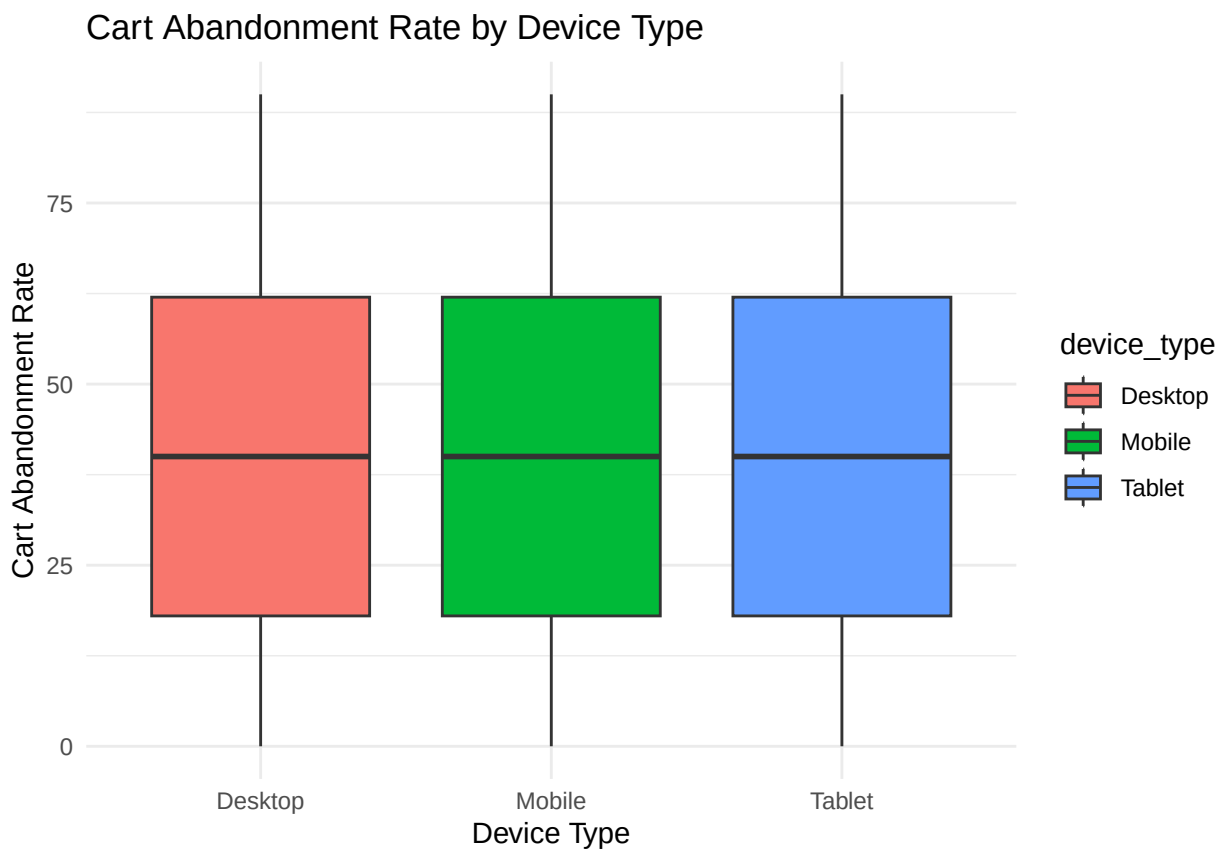
```
# ----------------------------------------------------------
# CLEAN CODE: Step 1 - Data Import and Preprocessing
# ----------------------------------------------------------

# Load required libraries for visualization and data manipulation [1]
library(ggplot2)
library(dplyr)

# 1. Import the E-commerce dataset

ecommerce_data <- read.csv("/Users/nguyenchinghia/Downloads/e_commerce_shopper_behaviour_and_lifestyle.c

# 2. Select the three key variables for the ANOVA model
analysis_data <- ecommerce_data %>%
  select(device_type, preferred_payment_method, cart_abandonment_rate) %>%
  na.omit() # Remove any missing values to ensure accurate analysis

# 3. Convert grouping variables to Factors (Mandatory for ANOVA in R)
```

```r
analysis_data$device_type <- as.factor(analysis_data$device_type)
analysis_data$preferred_payment_method <- as.factor(analysis_data$preferred_payment_method)

# Display the first 6 rows of the cleaned dataset
head(analysis_data)
```

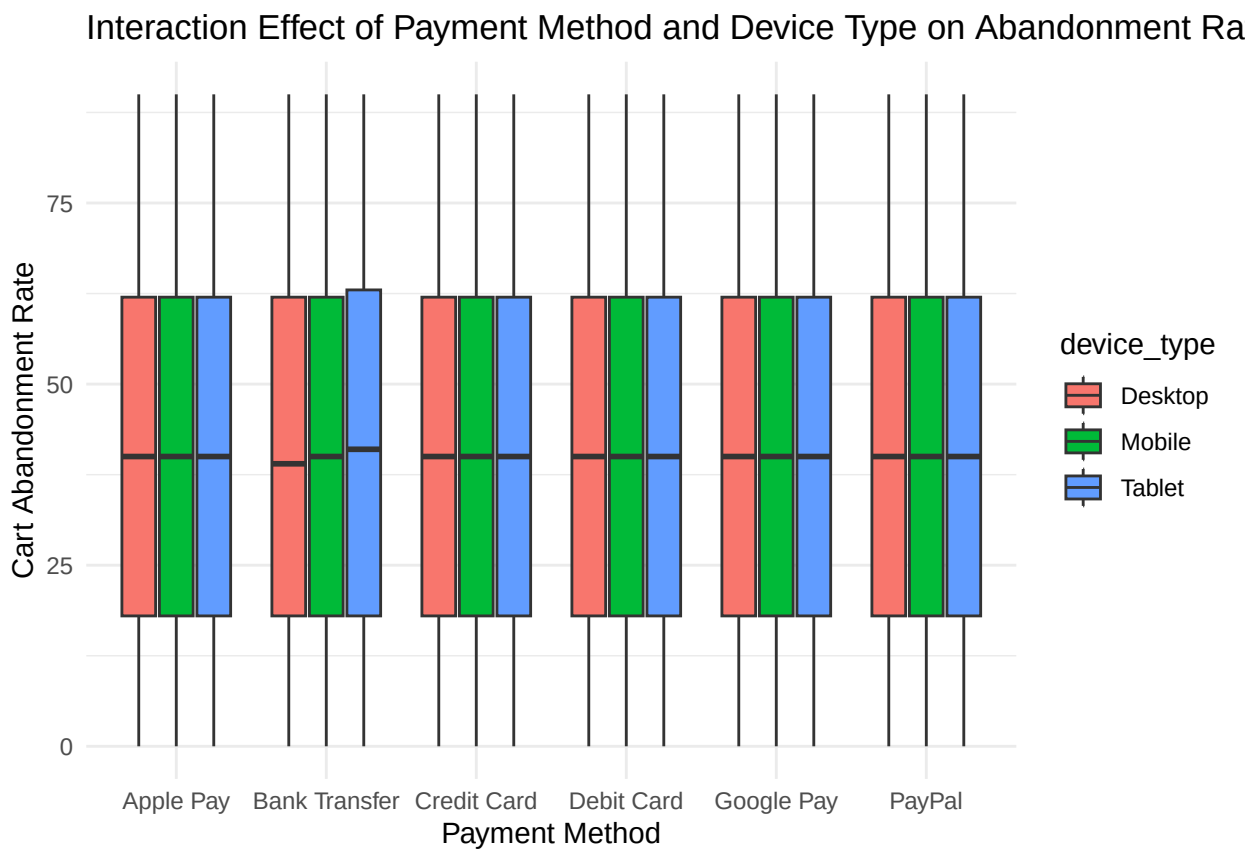
```
##   device_type preferred_payment_method cart_abandonment_rate
## 1      Mobile                   PayPal                     0
## 2      Mobile               Google Pay                    28
## 3      Mobile                   PayPal                    14
## 4     Desktop              Credit Card                    11
## 5     Desktop                Apple Pay                    57
## 6      Mobile                   PayPal                    46
```

### Data Visualization

```r
# ---------------------------------------------------------
# CLEAN CODE: Step 2 - Visualizing Experimental Data
# ---------------------------------------------------------

# Create a Boxplot to visually compare Cart Abandonment Rate by Device Type
ggplot(analysis_data, aes(x = device_type, y = cart_abandonment_rate, fill = device_type)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Cart Abandonment Rate by Device Type",
       x = "Device Type",
       y = "Cart Abandonment Rate")
```




```r
# Create an Interaction Plot to examine the combined effect of Payment Method and Device Type
ggplot(analysis_data, aes(x = preferred_payment_method, y = cart_abandonment_rate, fill = device_type))
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Interaction Effect of Payment Method and Device Type on Abandonment Rate",
       x = "Payment Method",
       y = "Cart Abandonment Rate")
```




**Statistical Analysis: Two-Way ANOVA**

```
# ---------------------------------------------------------
# CLEAN CODE: Step 3 - Running the Statistical Model
# ---------------------------------------------------------

# Build a Two-Way ANOVA model to assess main effects and interaction effects (using the * operator) [4]
anova_model <- aov(cart_abandonment_rate ~ device_type * preferred_payment_method, data = analysis_data

# Display the summary of the results (P-values, F-values) for causal inference [5]
summary(anova_model)
```

```
##                                        Df    Sum Sq Mean Sq F value Pr(>F)
## device_type                          2e+00       726   363.1   0.561 0.5704
## preferred_payment_method             5e+00      3164   632.8   0.978 0.4293
## device_type:preferred_payment_method 1e+01     11586  1158.6   1.791 0.0565 .
## Residuals                            1e+06 646838794   646.9
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 3. Findings & Implications

**1. Findings:**

- **Main Effect 1:** Device type alone does not significantly affect cart abandonment rate (p-value = 0.5704 > 0.05). In fact, customers abandon carts at similar rates across three given devices including

Mobile, Desktop, and Tablet platforms.

- **Main Effect 2:** The preferred payment method, as an independent factor, does not show a significant impact on cart abandonment (p-value = 0.4293 > 0.05).
- **Interaction Effect:** However, there is a **marginally significant interaction** between device type and payment method (p-value = 0.0565, approaching the 0.05 threshold). This suggests that certain combinations of devices and payment gateways may create friction during the checkout process.

**2. Implications:**

According to the causal inference from the ANOVA model, **E-commerce platforms should not treat devices or payment methods as isolated issues**. The marginal interaction effect strongly recommends that management should conduct targeted UI/UX testing on specific payment flows across different devices. For example, while a payment method might function flawlessly on a desktop, its interface might be cumbersome on a mobile screen, leading to a spike in abandonment.